SpecInfer

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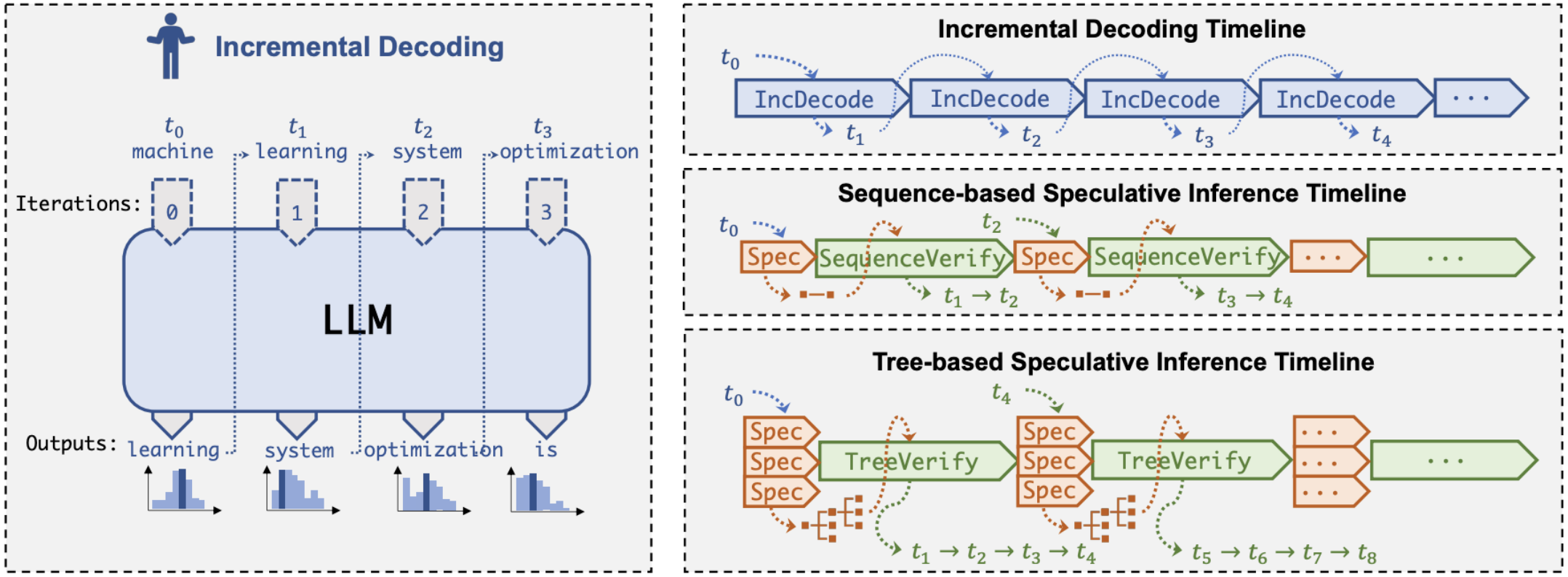

(a) Incremental decoding. **(b)** Timeline Comparison.

Figure 1. Comparing the incremental decoding approach used by existing LLM serving systems, the sequence-based speculative inference approach, and the tree-based speculative inference approach used by SpecInfer.

Fast Inference from Transformers via Speculative Decoding

Yaniv Leviathan [*1] **Matan Kalman** [*1] **Yossi Matias** [1]

Accelerating Large Language Model Decoding with Speculative Sampling

Charlie Chen[1], **Sebastian Borgeaud**[1], **Geoffrey Irving**[1], **Jean-Baptiste Lespiau**[1], **Laurent Sifre**[1] **and John Jumper**[1]

[1]All authors from DeepMind

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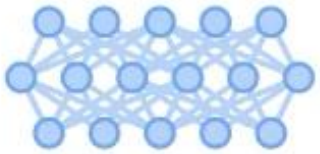
[START] japan ' s benchmark nikkei 225 index rose 226 . 69 points , or 1 . 5 percent , to 10 , 989 . 79 in late morning trading . [END]

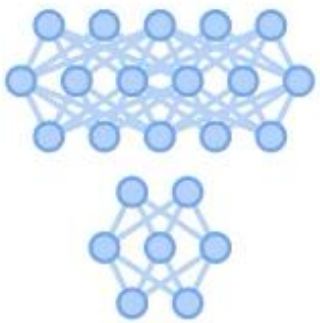
Figure 1. Our technique illustrated in the case of unconditional language modeling. Each line represents one iteration of the algorithm. The **green** tokens are the suggestions made by the approximation model (here, a GPT-like Transformer decoder with 6M parameters trained on lm1b with 8k tokens) that the target model (here, a GPT-like Transformer decoder with 97M parameters in the same setting) accepted, while the **red** and **blue** tokens are the rejected suggestions and their corrections, respectively. For example, in the first line the target model was run only once, and 5 tokens were generated.

WITHOUT SPECULATIVE DECODING



My favorite thing about fall

WITH SPECULATIVE DECODING



My favorite thing about fall

Algorithm 1 SpeculativeDecodingStep

Inputs: $M_p, M_q, prefix$.

▷ Sample γ guesses $x_{1,\ldots,\gamma}$ from M_q autoregressively.

for $i = 1$ **to** γ **do**

  $q_i(x) \leftarrow M_q(prefix + [x_1, \ldots, x_{i-1}])$

  $x_i \sim q_i(x)$

end for

▷ Run M_p in parallel.

$p_1(x), \ldots, p_{\gamma+1}(x) \leftarrow$

  $M_p(prefix), \ldots, M_p(prefix + [x_1, \ldots, x_\gamma])$

▷ Determine the number of accepted guesses n.

$r_1 \sim U(0,1), \ldots, r_\gamma \sim U(0,1)$

$n \leftarrow \min(\{i - 1 \mid 1 \leq i \leq \gamma, r_i > \frac{p_i(x)}{q_i(x)}\} \cup \{\gamma\})$

▷ Adjust the distribution from M_p if needed.

$p'(x) \leftarrow p_{n+1}(x)$

if $n < \gamma$ **then**

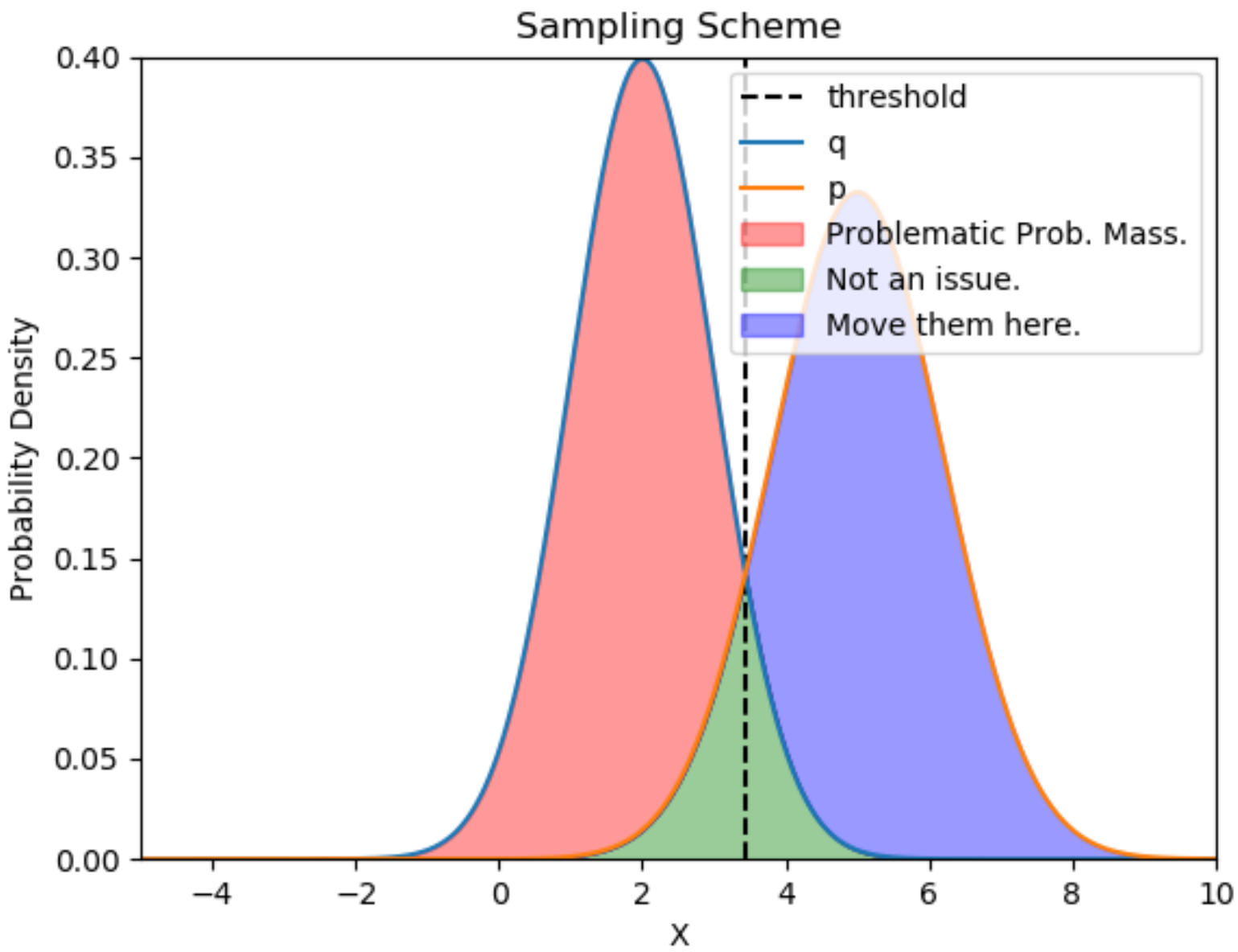
  $p'(x) \leftarrow norm(max(0, p_{n+1}(x) - q_{n+1}(x)))$

end if

▷ Return one token from M_p, and n tokens from M_q.

$t \sim p'(x)$

return $prefix + [x_1, \ldots, x_n, t]$



From “https://www.mlsys.ai/papers/speculative_decoding.html”

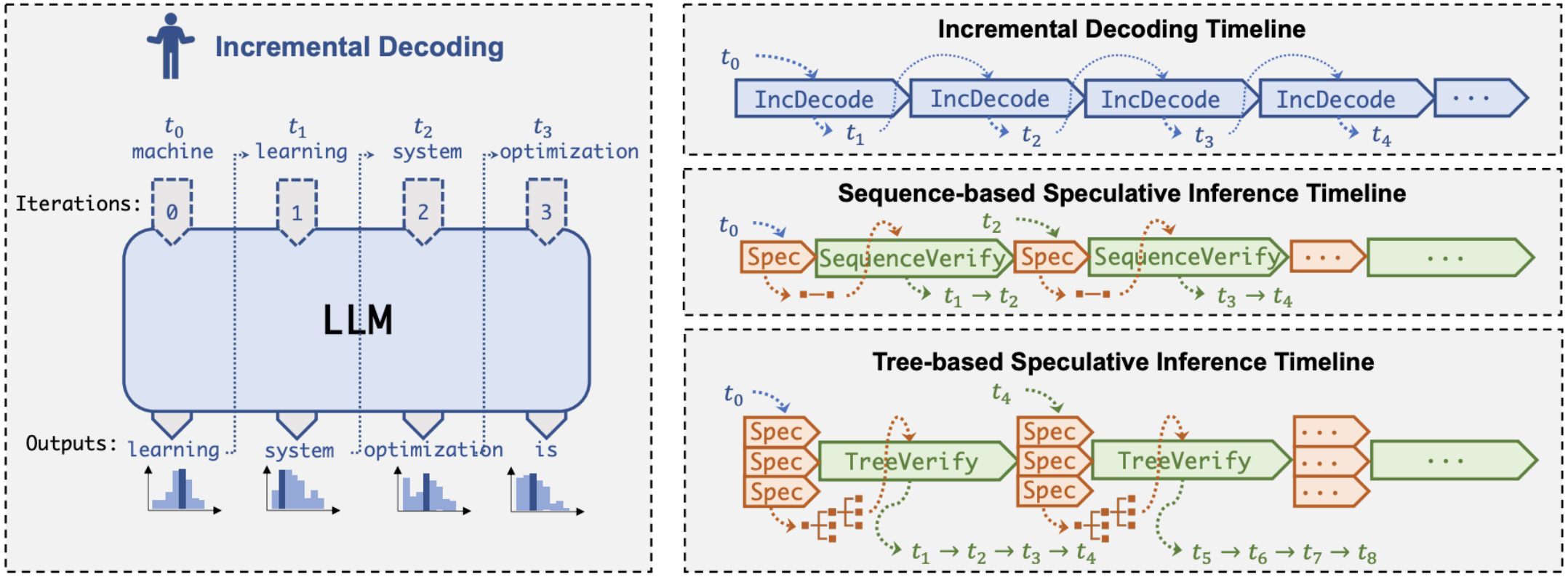


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Figure 1. Comparing the incremental decoding approach used by existing LLM serving systems, the sequence-based speculative inference approach, and the tree-based speculative inference approach used by SpecInfer.

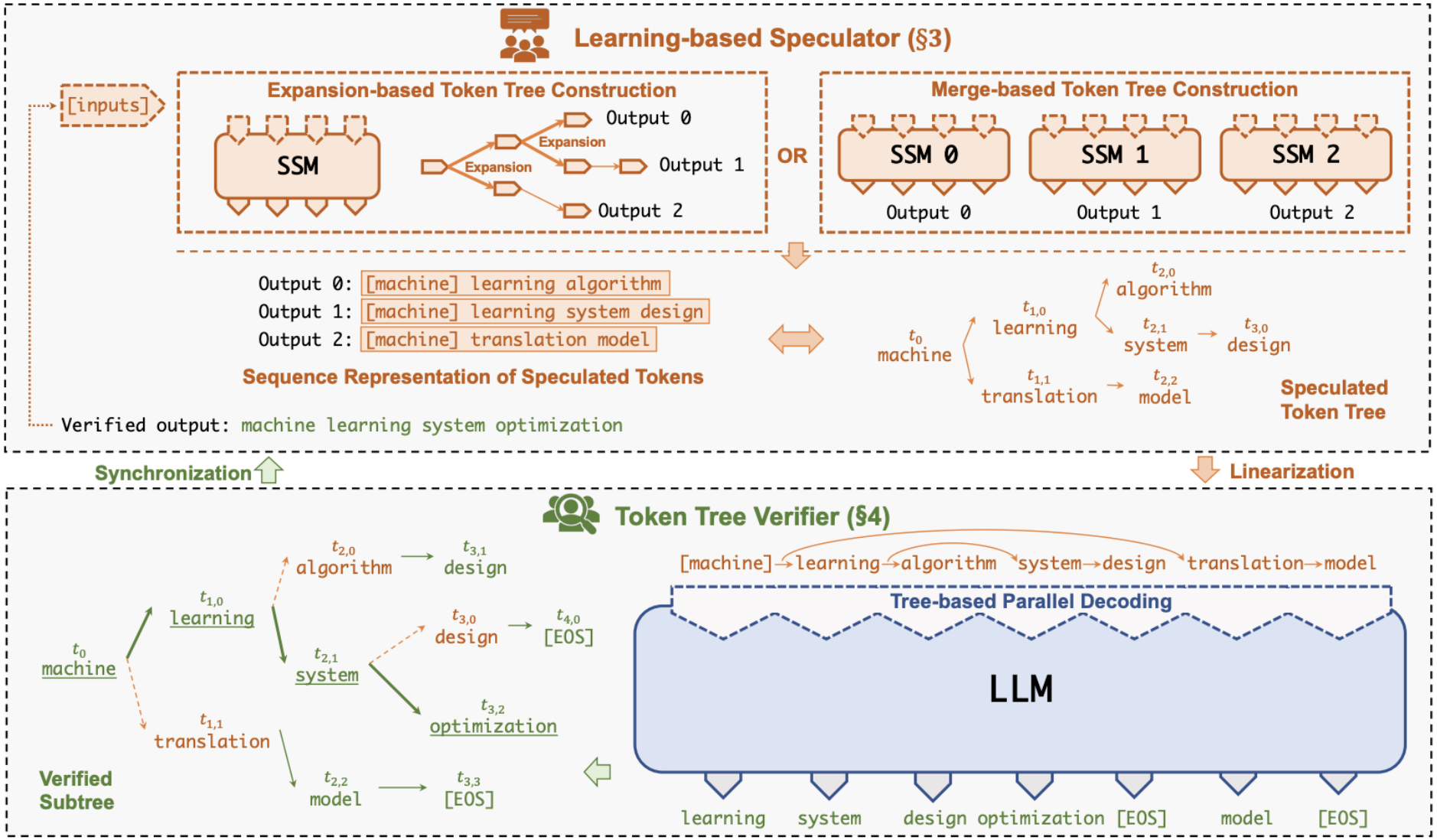


Figure 2. An overview of SpecInfer's tree-based speculative inference and verification mechanism.

Learning-based Speculator

- A key limitation of a single speculated sequence is that the probability of a successful alignment between the LLM and the speculated token sequence decays exponentially with the expected alignment length.

- Speculation only includes a single candidate token to verify per step.

- Solution: Encouraging more diverse speculated candidates per step, the probability of a successful match per step can be improved.
  - Using the top-5 tokens can increase the success rate from 70% to 89% for greedy decoding and from 57% to 97% for stochastic decoding.

Table 1. The success rate of verifying a token for LLaMA-7B using the top-k tokens derived from LLaMA-68M. The five prompt datasets are described in Section 6.1.

	Dataset	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
Greedy decoding	Alpaca	68%	77%	81%	84%	85%
	CP	69%	79%	83%	86%	87%
	WebQA	62%	72%	77%	80%	82%
	CIP	70%	81%	85%	88%	89%
	PIQA	63%	75%	79%	83%	85%
Stochastic decoding	Alpaca	54%	81%	91%	95%	97%
	CP	56%	82%	92%	95%	97%
	WebQA	52%	80%	90%	94%	96%
	CIP	57%	84%	92%	95%	97%
	PIQA	55%	82%	91%	95%	97%

How to expand the token tree?

- Naively, this means the tree is size of 5^(speculation length)

- SpecInfer uses fixed tree structure, e.g., (2,2,1)

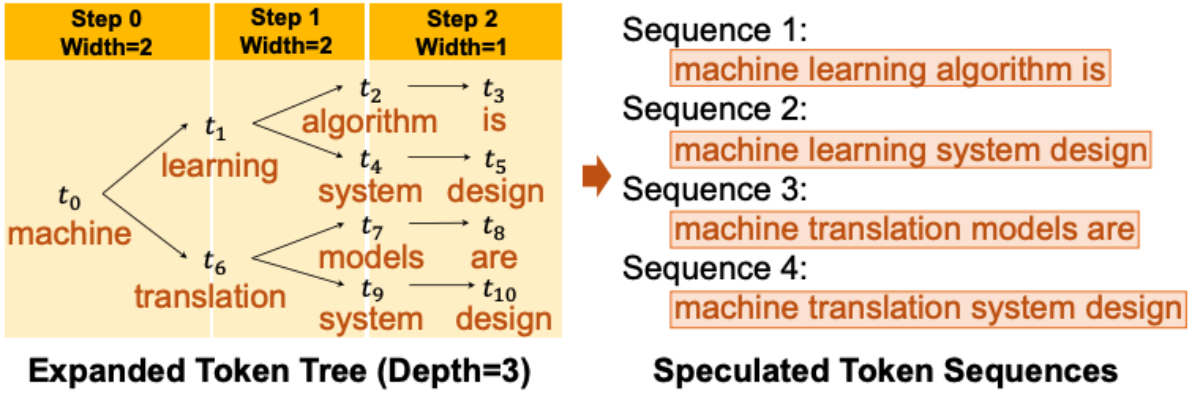


Figure 3. Illustration of token tree expansion.

Merged-based token tree construction

- SpecInfer uses an unsupervised method to collectively boost-tune a pool of SSMs to align their outputs with that of the LLM by leveraging adaptive boosting.

- Using a set of SSMs to cover a large text database

- During inference, SpecInfer runs multiple SSMs in parallel to construct token tree

Token Tree Verifier

- SpecInfer must address in token tree verification is efficiently computing the attention scores for all sequences of a token tree.

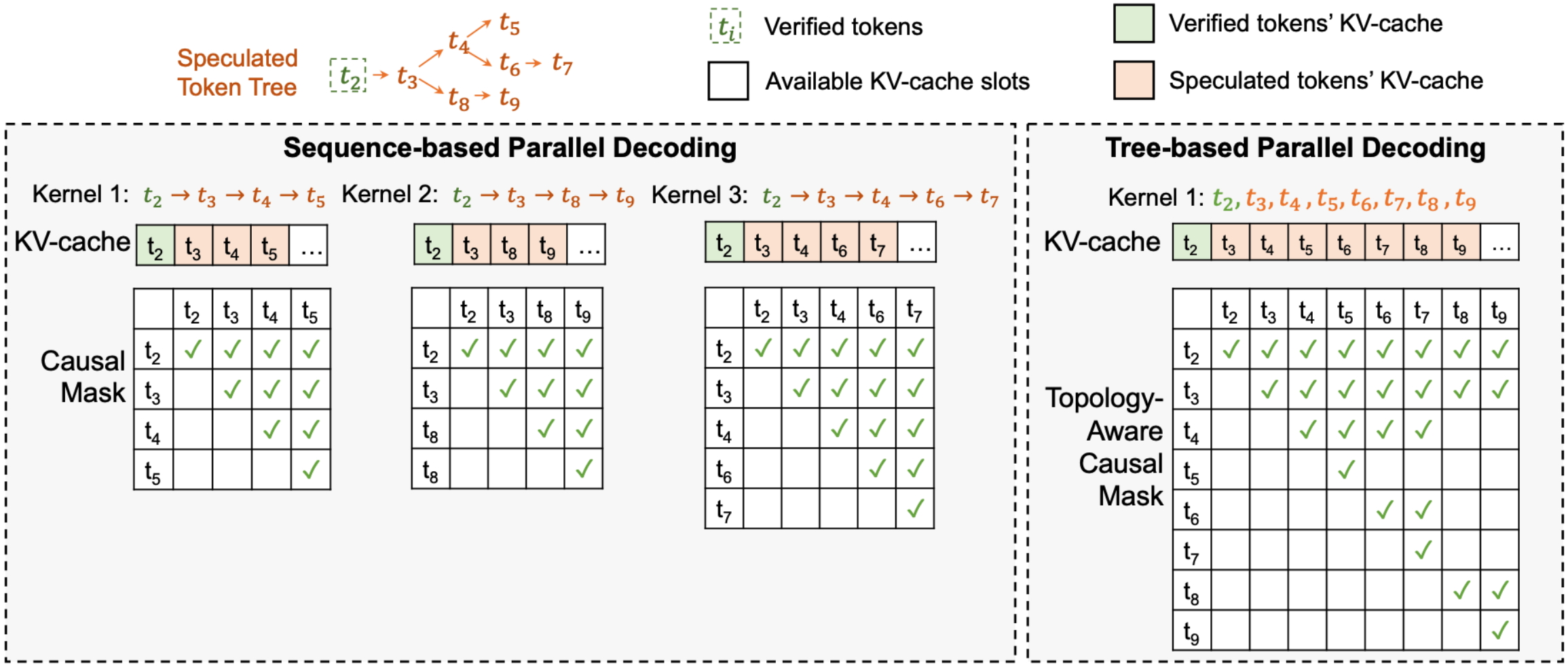
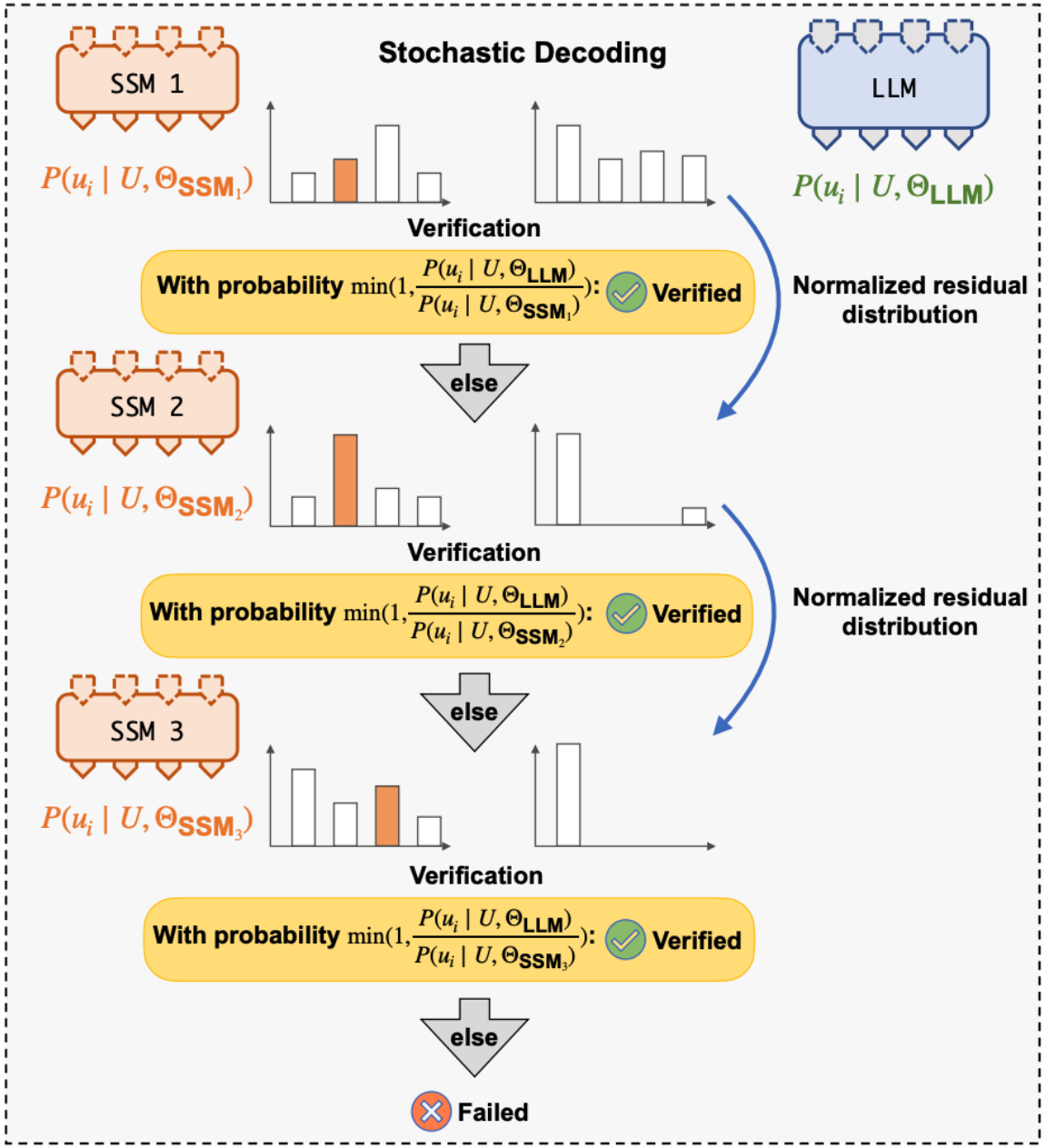


Figure 4. Comparing SpecInfer's tree-based parallel decoding with existing sequence-based decoding.

Token Tree Verifier

- SpecInfer must address in token tree verification is efficiently computing the attention scores for all sequences of a token tree.
  - Depth-first search to update key-value cache
  - Topology-aware casual mask
- Greedy decoding:
  - Check whether LLM's choice is in the token tree
- Stochastic decoding


Stochastic Decoding
SSM 1
LLM
$P(u_i \mid U, \Theta_{\text{SSM}_1})$
$P(u_i \mid U, \Theta_{\text{LLM}})$
Verification
With probability $\min(1, \frac{P(u_i \mid U, \Theta_{\text{LLM}})}{P(u_i \mid U, \Theta_{\text{SSM}_1})})$: Verified
Normalized residual distribution
else
SSM 2
$P(u_i \mid U, \Theta_{\text{SSM}_2})$
Verification
With probability $\min(1, \frac{P(u_i \mid U, \Theta_{\text{LLM}})}{P(u_i \mid U, \Theta_{\text{SSM}_2})})$: Verified
Normalized residual distribution
else
SSM 3
$P(u_i \mid U, \Theta_{\text{SSM}_3})$
Verification
With probability $\min(1, \frac{P(u_i \mid U, \Theta_{\text{LLM}})}{P(u_i \mid U, \Theta_{\text{SSM}_3})})$: Verified
else
Failed

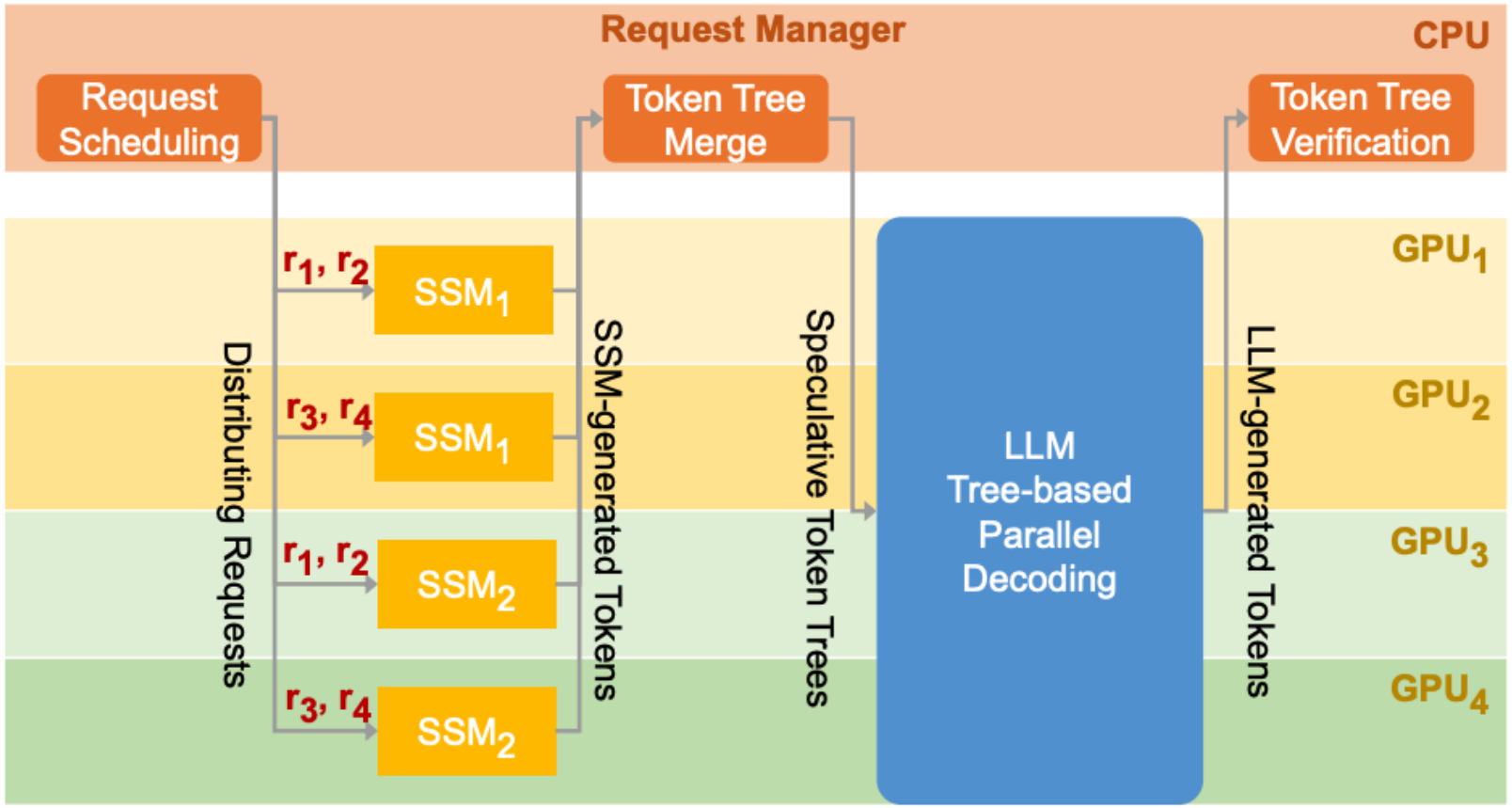
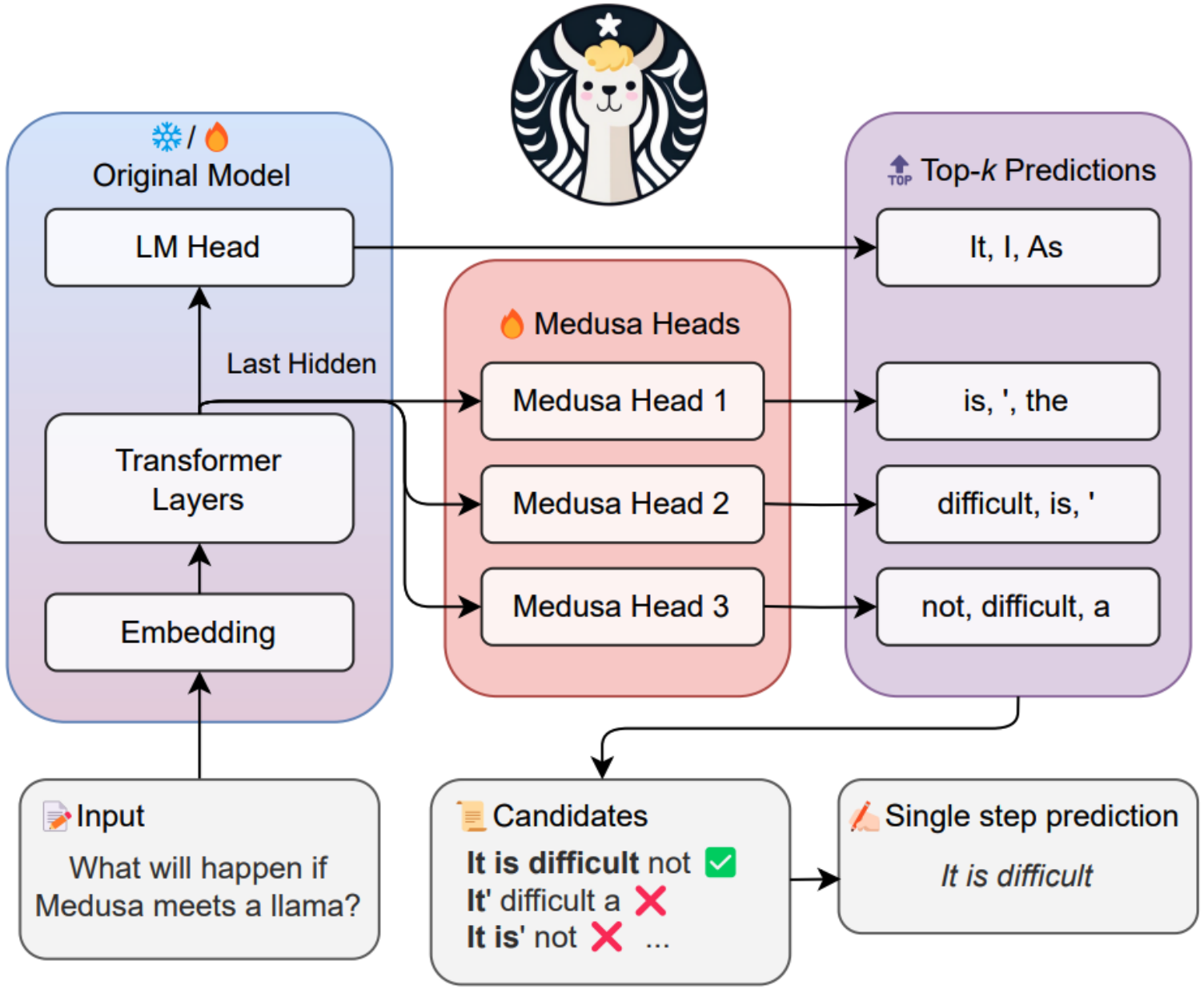


Figure 6. SpecInfer's workflow for one iteration of speculative inference and verification. SpecInfer uses data parallelism to serve SSMs, and combine tensor model parallelism and pipeline model parallelism for serving an LLM.

How about we directly train a draft model?

MEDUSA: Simple LLM Inference Acceleration Framework with Multiple Decoding Heads

Tianle Cai [*1 2] **Yuhong Li** [*3] **Zhengyang Geng** [4] **Hongwu Peng** [5] **Jason D. Lee** [1] **Deming Chen** [3] **Tri Dao** [1 2]


Original Model
LM Head
Last Hidden
Transformer Layers
Embedding
Medusa Heads
Medusa Head 1
Medusa Head 2
Medusa Head 3
Top-k Predictions
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is, ', the
difficult, is, '
not, difficult, a
Input
What will happen if Medusa meets a llama?
Candidates
It is difficult not
It' difficult a
It is' not ...
Single step prediction
It is difficult

How about we directly train a draft model?

EAGLE: Speculative Sampling Requires Rethinking Feature Uncertainty

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[♠]Peking University [‡]Microsoft Research [♣]University of Waterloo [†]Vector Institute
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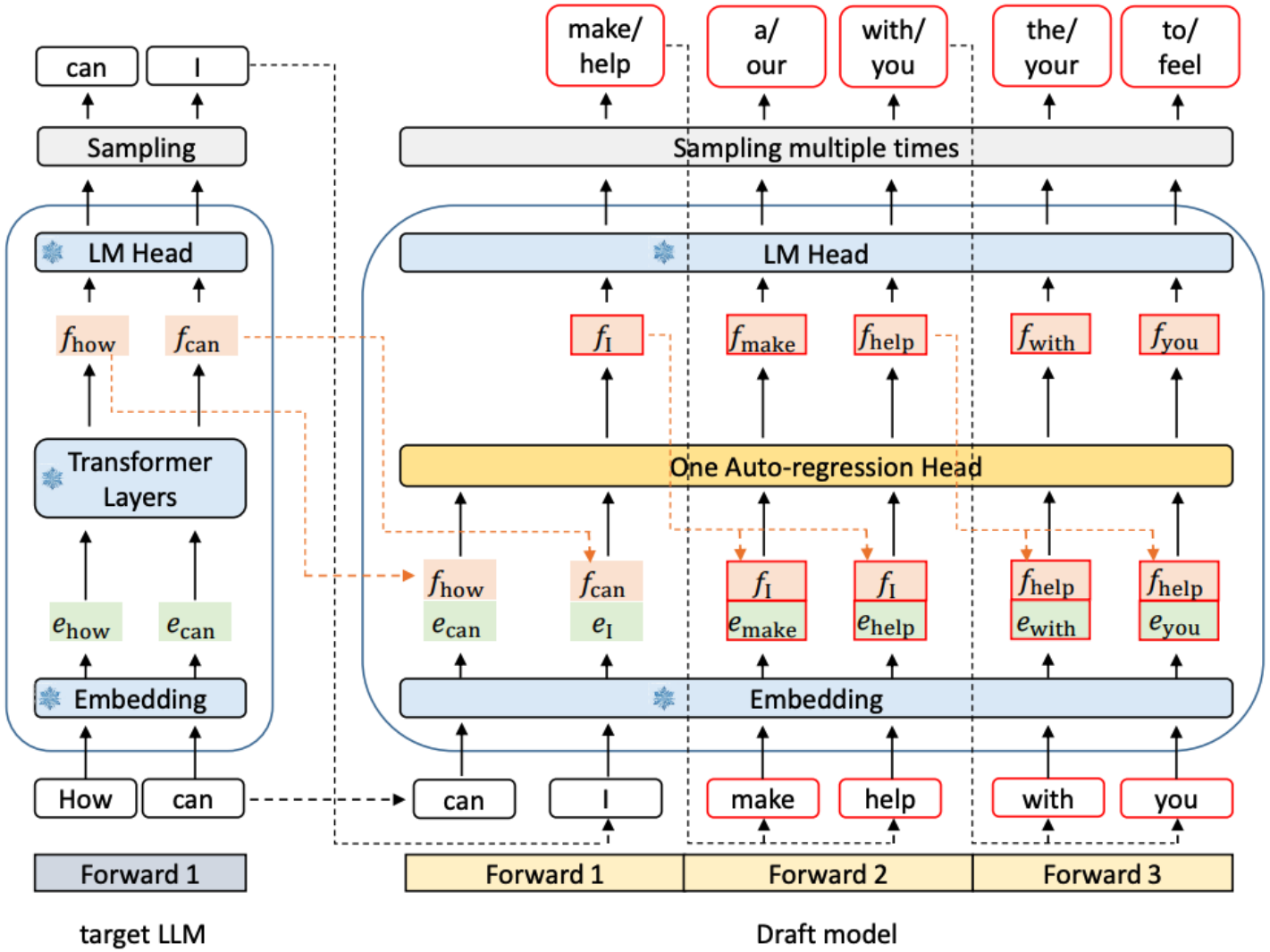
https://github.com/SafeAILab/EAGLE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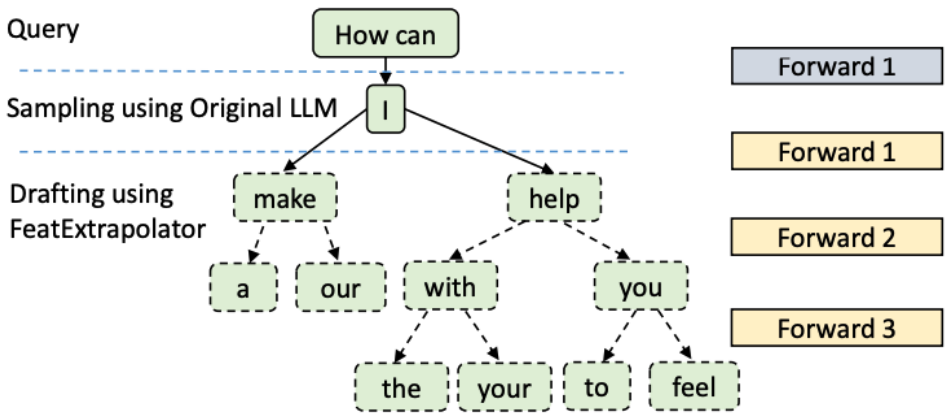
Draft model


Query
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Drafting using FeatExtrapolator
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Forward 1
Forward 1
Forward 2
Forward 3